

Question 3 - Accessibility: Smart Navigation Cane

The Problem

Blind and low-vision Pedestrians have a genuinely hard time getting around unfamiliar streets safely. A regular white cane only picks up obstacles within arm's reach, and gives zero context beyond that. It can't tell you whether a crossing signal is green, or that scaffolding is blocking the path ahead.

System Overview

Who it's for:

blind and visually impaired Pedestrians in cities, and potentially older users with declining eyesight.

Input:

Ultrasonic/lidar sensors plus a small forward-facing camera for obstacles and signage, a mic on a connected earpiece for spoken question and GPS for location.

Output:

directional vibration in handle

Main interaction style:

Mostly Ubiquitous Computing, Since it senses the environment constantly in the background combined with agent based interaction. it guides or answers like a Personal Assistant

Why it's different:

a normal cane is short range, and a phone app makes you look at a screen, pulling attention off your surrounding. This system stays hands free, keeps hearing open, and reads the situation as you go instead of just pointing towards a destination

Privacy / Ethical Risks

- The camera record the street constantly, including bystander's face. a real risk for surveillance or data leaks
- The Obstacle recognition AI could be worse in places it wasn't well trained on, giving some users less reliable, less safe performance with them knowing why

Technology Limitations:

Sensors will be less reliable in heavy rain, fog or low light - exactly when the users need them most, risking the device dying mid-trip

Accessibility

- Voice and vibration are default output
- Power and volume controls are big and tactile, usable without looking
- Setup is started through by voice
- A vibration mode also covers users that are hard of hearing